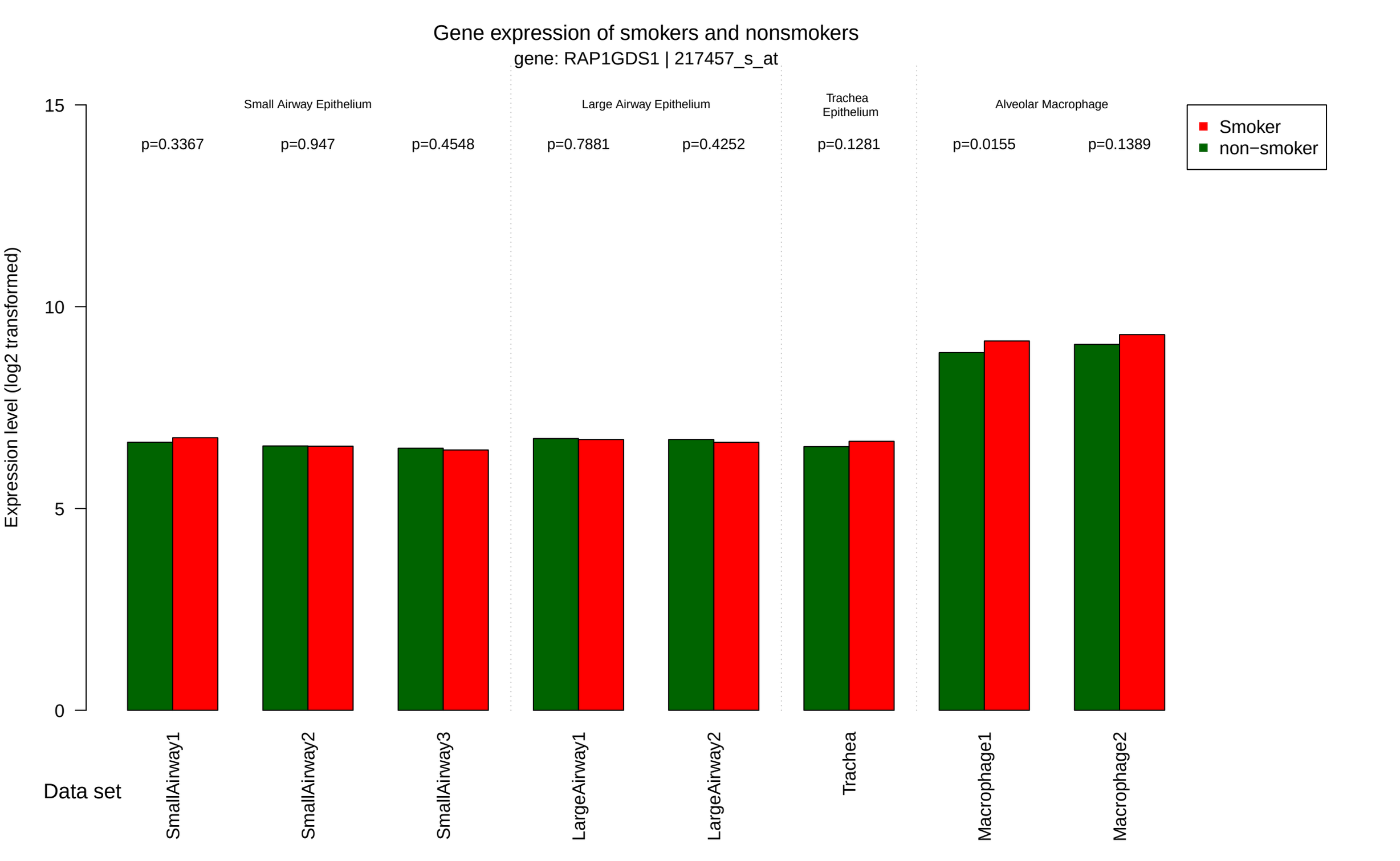
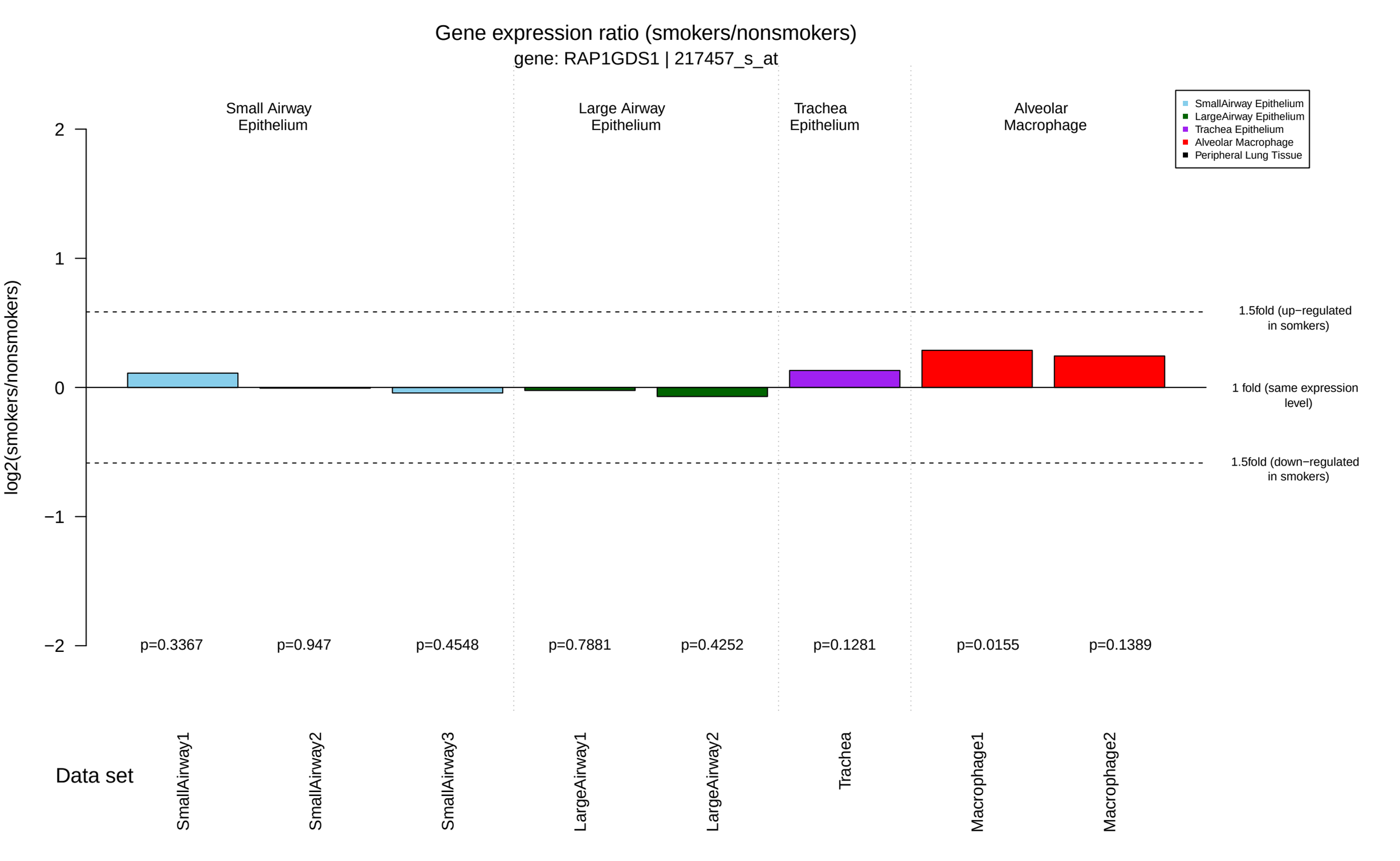
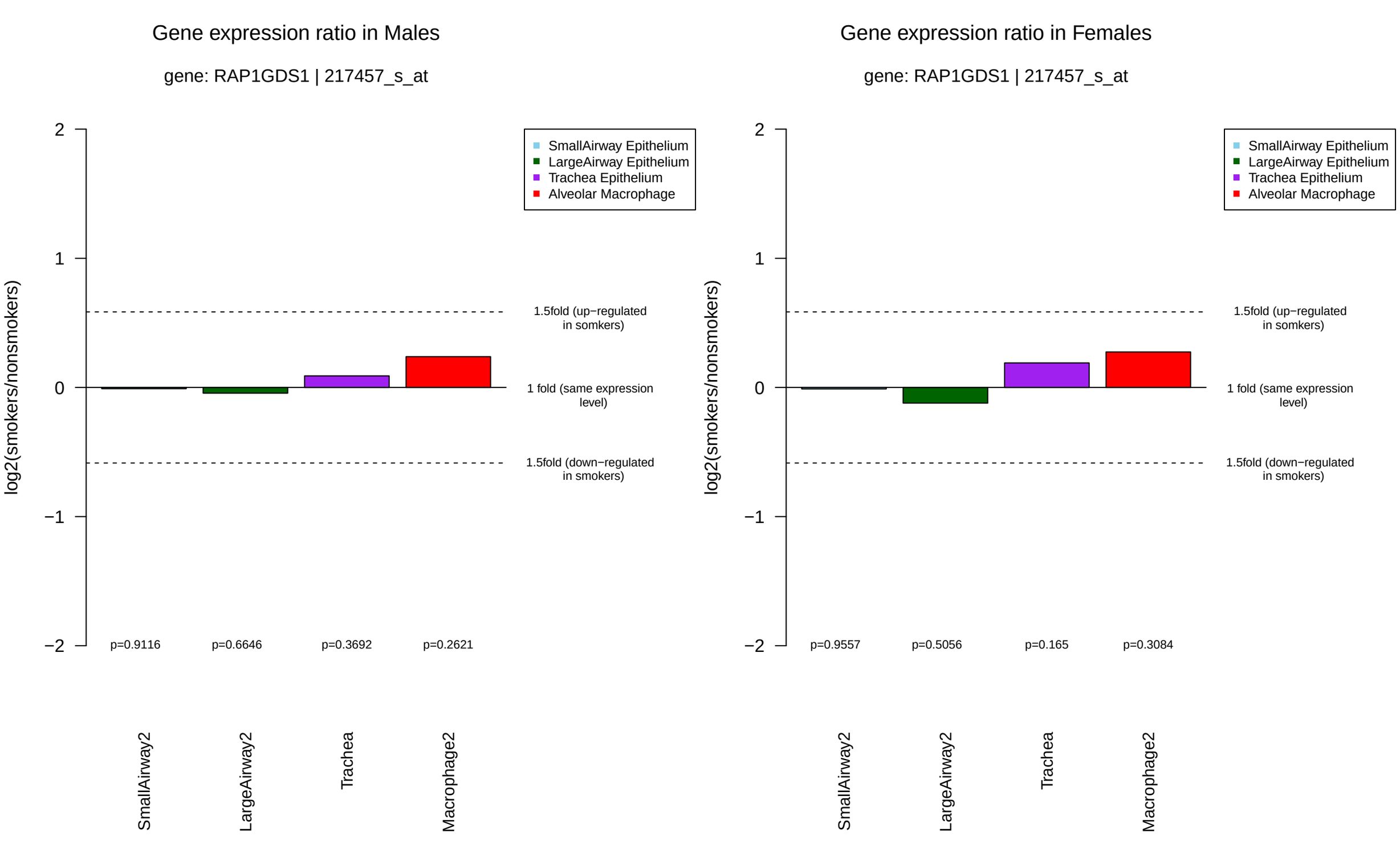
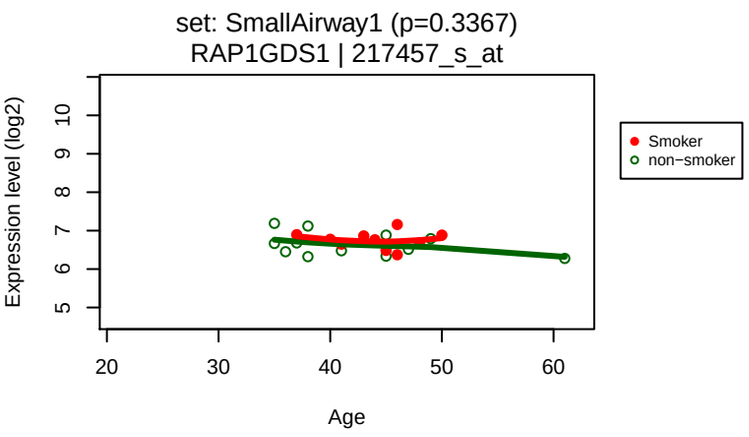
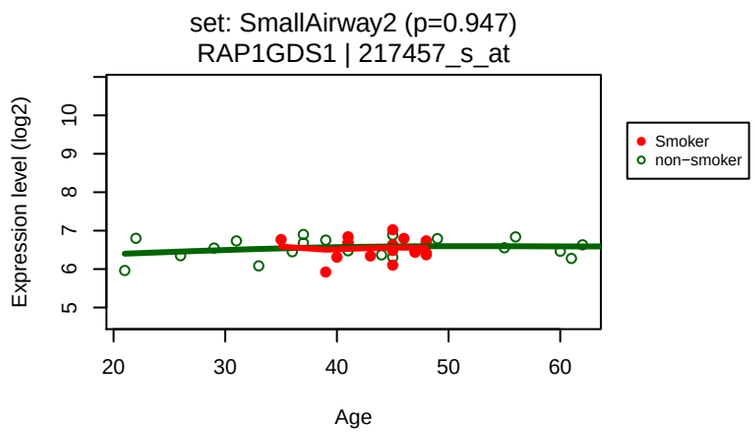
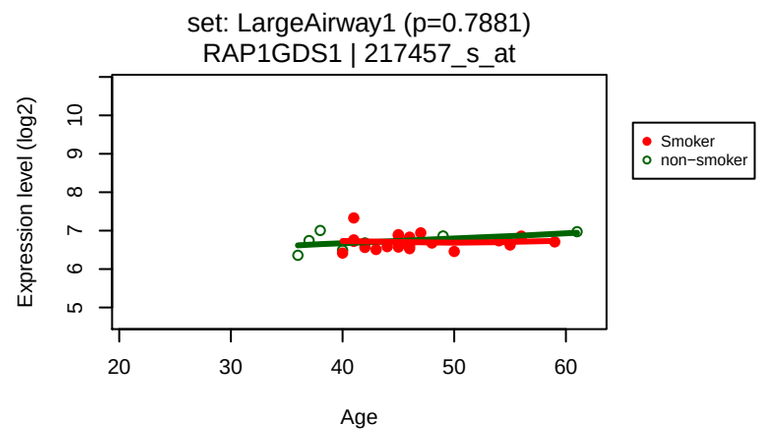

Gene expression of smokers and nonsmokers
gene: RAP1GDS1 | 217457_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3367
p=0.947
p=0.4548
p=0.7881
p=0.4252
p=0.1281
p=0.0155
p=0.1389
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

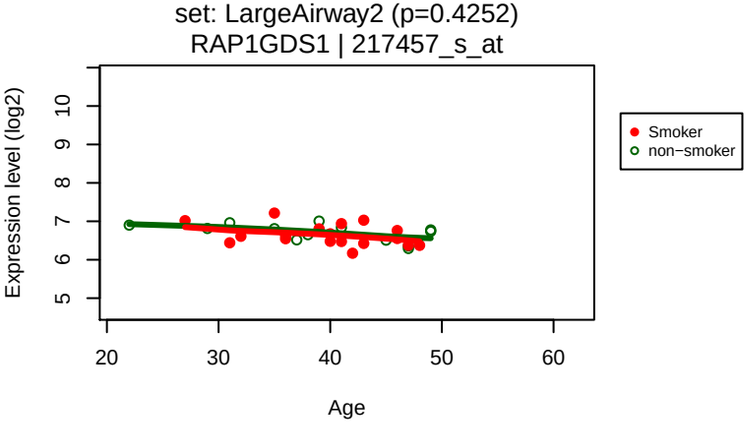

Gene expression ratio (smokers/nonsmokers)
gene: RAP1GDS1 | 217457_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3367
p=0.947
p=0.4548
p=0.7881
p=0.4252
p=0.1281
p=0.0155
p=0.1389
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

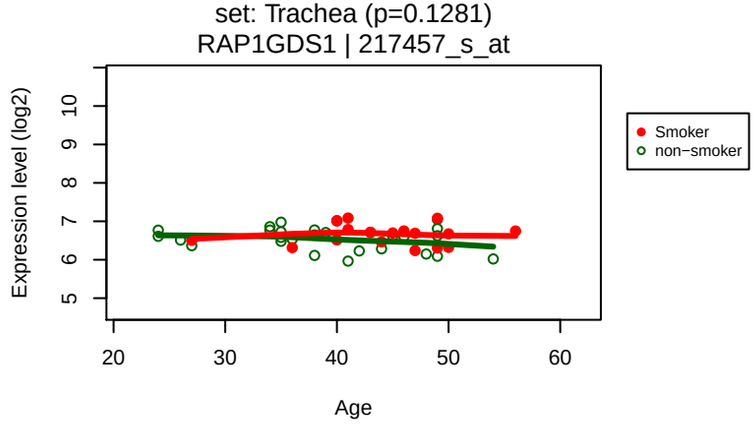

Gene expression ratio in Males
gene: RAP1GDS1 | 217457_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9116
p=0.6646
p=0.3692
p=0.2621
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP1GDS1 | 217457_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9557
p=0.5056
p=0.165
p=0.3084
SmallAirway2
LargeAirway2
Trachea
Macrophage2

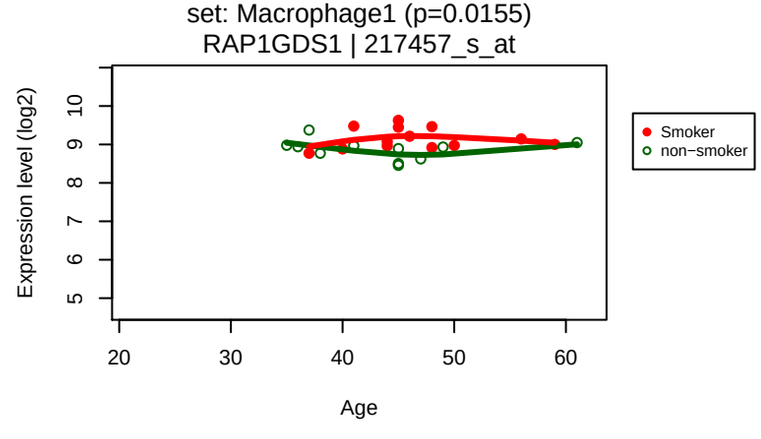

set: SmallAirway1 (p=0.3367)
RAP1GDS1 | 217457_s_at
Expression level (log2)
Age
Smoker
non−smoker

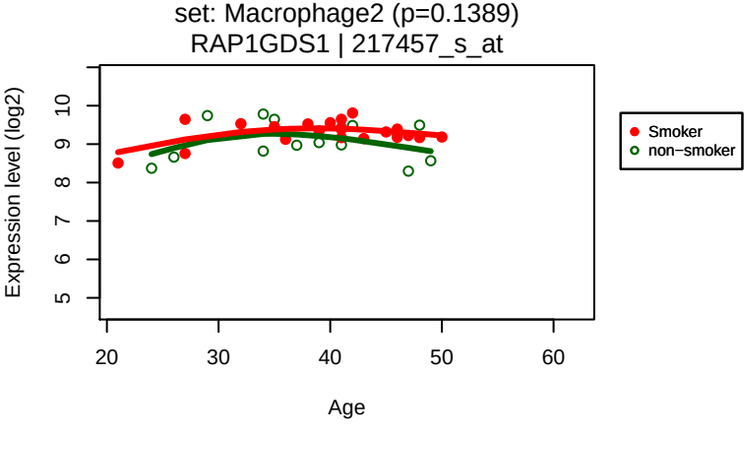

set: SmallAirway2 (p=0.947)
RAP1GDS1 | 217457_s_at
Expression level (log2)
Age
Smoker
non−smoker

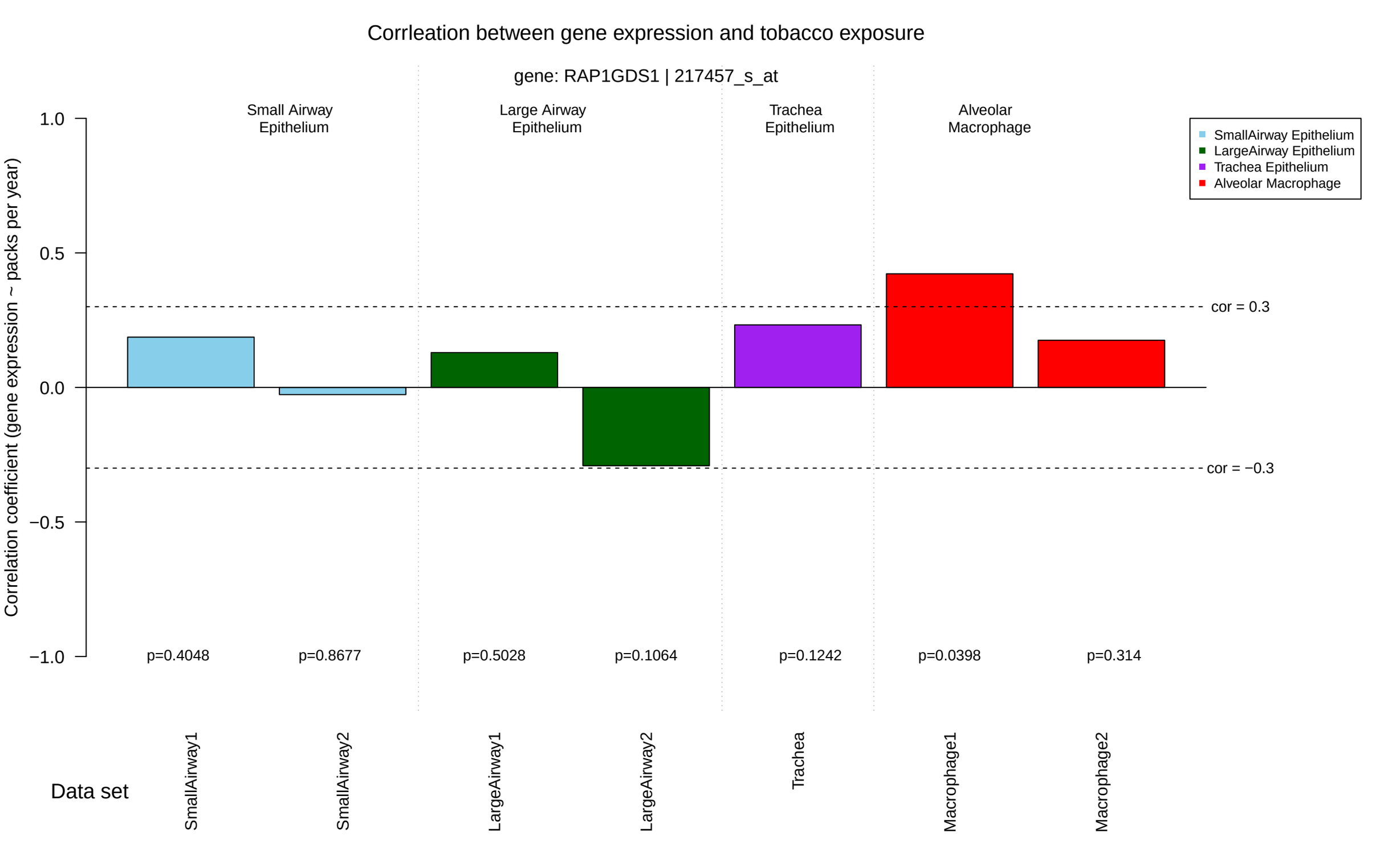

set: LargeAirway1 (p=0.7881)
RAP1GDS1 | 217457_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4252)
RAP1GDS1 | 217457_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1281)
RAP1GDS1 | 217457_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0155)
RAP1GDS1 | 217457_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1389)
RAP1GDS1 | 217457_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP1GDS1 | 217457_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4048
p=0.8677
p=0.5028
p=0.1064
p=0.1242
p=0.0398
p=0.314
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2